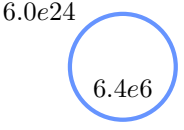


$$G = 6.7e-11$$

$$F = \frac{Gm_1m_2}{{d_{12}}^2}$$

$$\frac{Gm_{_S}m_{_E}}{d_{_S}d_{_S}} = \frac{6.7e-11 \cdot 2.0e30 \cdot 6.0e24}{1.5e11 \cdot 1.5e11} = 3.6e22\,[kg\,m/s^2]$$

$$\frac{Gm_{_S}}{d_{_S}d_{_S}} = \frac{6.7e-11 \cdot 2.0e30}{1.5e11 \cdot 1.5e11} = 6.0e-3\,[m/s^2]$$





$$\frac{Gm_{_M}m_{_E}}{d_{_M}d_{_M}} = \frac{6.7e-11 \cdot 7.4e22 \cdot 6.0e24}{3.8e8 \cdot 3.8e8} = 2.1e20\,[kg\,m/s^2]$$

$$\frac{Gm_{_M}}{d_{_M}d_{_M}} = \frac{6.7e-11 \cdot 7.4e22}{3.8e8 \cdot 3.8e8} = 3.4e{-}5\,[m/s^2]$$